Matrices: Introduction

Questions 1–79

Question 1. Write down the dimensions of each of these matrices:

a) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$

b) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

c) $\begin{pmatrix} 3 & 2 & -8 \end{pmatrix}$

d) $\begin{pmatrix} -3 & 2 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 4 & 3 & -9 \\ 17 & 0 & 10 \\ 2 & 1 & 25 \end{pmatrix}$

f) $\begin{pmatrix} 13 & 21 \end{pmatrix}$

g) $\begin{pmatrix} 5 \\ -4 \\ -99 \end{pmatrix}$

h) $\begin{pmatrix} -1 & 3 \\ -3 & 2 \\ -8 & 1 \end{pmatrix}$

i) $\begin{pmatrix} 3 & 4 & 4 & 2 & 1 & 0 \\ 6 & 0 & 3 & 8 & 2 & 2 \\ 1 & 0 & 2 & 9 & 9 & 1 \\ 2 & 0 & 6 & 0 & 5 & 5 \\ 0 & 5 & 5 & 7 & 3 & 4 \end{pmatrix}$

Question 2. Write down an example of a matrix with the following dimensions:

a) 2×3

b) 3×1

c) 4×2

d) 3×3

e) 1×4

f) 5×3

g) 2×5

Question 3. Find:

a) $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 5 & 2 \\ -1 & 7 \end{pmatrix}$

b) $\begin{pmatrix} 6 & 2 \\ 3 & -1 \end{pmatrix} - \begin{pmatrix} 4 & -7 \\ 6 & 9 \end{pmatrix}$

c) $\begin{pmatrix} 8 & -3 & 1 \\ 2 & -9 & 3 \end{pmatrix} + \begin{pmatrix} 3 & 2 & -4 \\ -9 & -1 & 3 \end{pmatrix}$

d) $\begin{pmatrix} -3 & 10 \\ 2 & -6 \\ 9 & -2 \end{pmatrix} - \begin{pmatrix} 6 & -4 \\ -8 & -8 \\ -1 & 2 \end{pmatrix}$

Question 4. Find:

a) $2\begin{pmatrix} 1 & -3 & -9 \\ 5 & 2 & 4 \end{pmatrix}$

b) $-5\begin{pmatrix} 3 & 2 & -1 \\ -6 & 5 & 2 \\ -2 & 3 & -12 \end{pmatrix}$

Question 5. Find:

a) $\begin{pmatrix}5\\7\end{pmatrix}+2\begin{pmatrix}-3\\4\end{pmatrix}$

b) $3\begin{pmatrix}3&-2\\1&4\end{pmatrix}-\begin{pmatrix}5&2\\9&1\end{pmatrix}$

c) $\frac{3}{2}\begin{pmatrix}8&4\\20&3\\-3&2\end{pmatrix}-4\begin{pmatrix}-2&3\\-1&2\\8&\frac{1}{2}\end{pmatrix}$

Question 6. Given that $\mathbf{A}=\begin{pmatrix}7&2\\-1&3\end{pmatrix}$, $\mathbf{B}=\begin{pmatrix}2&0\\1&-2\end{pmatrix}$ and $\mathbf{C}=\begin{pmatrix}4&20\\9&-11\end{pmatrix}$, find:

a) $\mathbf{A}+2\mathbf{B}+3\mathbf{C}$

b) $\mathbf{B}-\mathbf{A}-\mathbf{C}$

c) $\mathbf{C}-2(\mathbf{A}+\mathbf{B})$

Question 7. Find the value of a and b given $\begin{pmatrix} a & 2 \\ 3 & b \end{pmatrix} + \begin{pmatrix} b & 4 \\ 2 & 2a \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix}$.

Question 8. Find the value of k and m given $\begin{pmatrix} 4 & k \\ k & m \\ 6 & k \end{pmatrix} - \begin{pmatrix} 2m & 3k \\ m & 5k \\ 2 & m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$.

Question 9. Find the value of p and q given $\begin{pmatrix} 2 & 5q \\ 8 & 3p \\ p & p \end{pmatrix} + q \begin{pmatrix} 2q & 4 \\ p & q \\ 3q & 2 \end{pmatrix} = \begin{pmatrix} 4 & -p \\ q & 28 \\ 12 & 7 \end{pmatrix}$.

Question 10. Given $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 & 2 & 3 \\ 1 & 9 & 0 \end{pmatrix}$, explain why we cannot find $\mathbf{A} + \mathbf{B}$.

Question 11. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$.

a) Show that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under addition.

b) Use the matrices $\mathbf{A}$ and $\mathbf{B}$ to show that matrix subtraction is not, in general, commutative.

Question 12. Given $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}$:

a) Show that the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under addition.

b) Use the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ to show that matrix subtraction is not, in general, associative.

Question 13. Given $\mathbf{A} = \begin{pmatrix} 2(x+1) & 3(1-x) \\ -3(2-x) & 4(5-x) \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8x+1 & 5x-2 \\ 2-x & 3+x \end{pmatrix}$, find $\mathbf{A} - \mathbf{B}$ and write it in its simplest form.

Question 14. Given $\mathbf{A} = \begin{pmatrix} 7k+3 & 3+k \\ 5k+1 & 9-6k \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4k-3 & 13+6k \\ 5+7k & 1-10k \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 3 & -5 \\ -2 & 4 \end{pmatrix}$, for constant k, show that $\mathbf{A} - \mathbf{B} = f(k)\mathbf{C}$, where f is a function of k to be found.

Question 15. Write down the following 2×2 matrices:

a) $\mathbf{I}$

b) $3\mathbf{I}$

c) $\mathbf{0}$

Question 16. Write down the following 3×3 matrices:

a) $\mathbf{I}$

b) $-2\mathbf{I}$

c) $\mathbf{0}$

Question 17. Given $\mathbf{A} = \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix}$ and $\mathbf{I}$ and $\mathbf{0}$ are the 2×2 identity and zero matrices respectively, find:

a) $\mathbf{0} - \mathbf{A}$
b) $\mathbf{A} + \mathbf{I}$
c) $2\mathbf{I} - 3\mathbf{A}$
d) $\mathbf{I} - (2\mathbf{A} + \mathbf{0})$
e) $\mathbf{0} - (\mathbf{I} - 3\mathbf{A})$
f) $10\mathbf{A} - (6\mathbf{I} - 5\mathbf{A})$

Question 18. Find the values of a, b and c given $4\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 3 \end{pmatrix} - \begin{pmatrix} b & 0 & 0 \\ 0 & 2c & 0 \\ 0 & 0 & a \end{pmatrix} = 2\mathbf{I}$.

Question 19. Find the values of p, q and r given $2\begin{pmatrix} p & q \\ 3 & r \end{pmatrix} + k\mathbf{I} = 4\begin{pmatrix} 2q & r \\ p & 2p \end{pmatrix}$.

Question 20. It is given that $\mathbf{A}$ is a 3×2 matrix and $\mathbf{B}$ is a 2×2 matrix. Explain why only one of $\mathbf{AB}$ and $\mathbf{BA}$ can be calculated.

Question 21. It is given that $\mathbf{A}$ is a 3×3 matrix and $\mathbf{B}$ is a 3×4 matrix. Explain why only one of $\mathbf{AB}$ and $\mathbf{BA}$ can be calculated.

Question 22. It is given that $\mathbf{A}$ is a 3×3 matrix and $\mathbf{B}$ is a 2×3 matrix. Explain why only one of $\mathbf{A}^2$ and $\mathbf{B}^2$ can be calculated.

Question 23. Six matrices, **A**, **B**, **C**, **D**, **E** and **F** have the following dimensions: **A** : 2×2, **B** : 2×4, **C** : 3×4, **D** : 3×2, **E** : 3×1, **F** : 4×2. In each case, write down the dimensions of the given product, or state that the matrix cannot be calculated:

a) **AB**
b) **BC**
c) **DA**
d) **DE**
e) **ED**
f) **FB**
g) **CF**
h) **BA**

Question 24. Find:

a) $\begin{pmatrix} 2 & 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$

b) $\begin{pmatrix} -3 & 1 & -2 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -4 \\ 3 \\ 1 \end{pmatrix}$

c) $\begin{pmatrix} 4 & 3 \\ -1 & 2 \\ 5 & -6 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix}$

d) $\begin{pmatrix} 3 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix}$

e) $\begin{pmatrix} 5 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 2 \\ 1 & 1 \end{pmatrix}$

f) $\begin{pmatrix} 3 & 4 & 1 \\ 2 & 3 & 0 \\ 0 & 3 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}$

g) $\begin{pmatrix} 3 & 2 & -5 \end{pmatrix} \begin{pmatrix} -1 & -2 \\ 2 & 6 \\ 9 & 0 \end{pmatrix}$

h) $\begin{pmatrix} 4 & 2 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} 5 & 2 & -9 \\ -3 & 1 & 10 \end{pmatrix}$

i) $\begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 & -8 \end{pmatrix}$

j) $\begin{pmatrix} a \\ b \\ c \end{pmatrix} \begin{pmatrix} b & c & a \end{pmatrix}$, where a, b and c are constants.

Question 25. Given $\mathbf{A} = \begin{pmatrix} 5 & 2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$, find $\mathbf{ABA}$.

Question 26. Find:

a) $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}^2$

c) $\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 5 \end{pmatrix}$

d) $\begin{pmatrix} -1 & 2 \\ -3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ -4 & 2 \end{pmatrix}$

e) $\begin{pmatrix} 8 & -3 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} -6 & 5 \\ 3 & -5 \end{pmatrix}$

f) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^3$

g) $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}^3$

h) $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$

Question 27. Given $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 3 \\ 2 & 5 \end{pmatrix}$, find $\mathbf{A}^2 - \mathbf{B}^2$.

Question 28. Given $\mathbf{A} = \begin{pmatrix} 1 & -3 \\ 8 & 1 \end{pmatrix}$ and $(\mathbf{A} - \mathbf{I})^4 = k\mathbf{I}$, find the value of k.

Question 29. Find the value of k given that $\begin{pmatrix} k & 0 & 1 \\ 1 & k & 2 \end{pmatrix} \begin{pmatrix} k & 1 \\ 2 & k \\ 1 & k \end{pmatrix} = \begin{pmatrix} 1-2i & 2-2i \\ 5-3i & 3-4i \end{pmatrix}$.

Question 30. It is given that $\mathbf{A} = \begin{pmatrix} 4 & -i \\ 3 & 2i \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} k & -3i \\ 4i & -2 \end{pmatrix}$, where k is a constant.

a) Find $\mathbf{AB}$.
b) Find $\mathbf{BA}$.
c) Explain why there is no value of k such that $\mathbf{AB} = \mathbf{BA}$.

Question 31. It is given that $\mathbf{A} = \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix}$. Find all possible products of two of these three matrices.

Question 32. Find:

a) $\begin{pmatrix} a & 2 \\ 2a & 1 \end{pmatrix} \begin{pmatrix} 3 & a \\ 4 & 1 \end{pmatrix}$

b) $\begin{pmatrix} 2b & b \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3b \\ 2 & 2b \end{pmatrix}$

Question 33. For a constant k, $\mathbf{A} = \begin{pmatrix} -2 & k & 0 \\ 1 & k & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & k-1 \\ -1 & 2k \end{pmatrix}$. Given that $\mathbf{AB}$ is a symmetric matrix, find the possible values of k, and hence find the possible matrices $\mathbf{AB}$.

Question 34. Given $\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix}$, find:

a) $\mathbf{AB}$
b) $\mathbf{BA}$

Question 35.

a) Show that $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^3 = \begin{pmatrix} a^3 & 0 & 0 \\ 0 & b^3 & 0 \\ 0 & 0 & c^3 \end{pmatrix}$.

b) Find the matrix $\mathbf{A}^3$ given $\mathbf{A} = \begin{pmatrix} \sin\theta & 0 & 0 \\ 0 & -\cos\theta & 0 \\ 0 & 0 & -\sin\theta \end{pmatrix}$.

Question 36. Find the matrix $\mathbf{A}$ such that $\begin{pmatrix} 2 & 3 \\ -3 & 4 \end{pmatrix}\mathbf{A} = \begin{pmatrix} 12 \\ -69 \end{pmatrix}$.

Question 37. The matrix $\mathbf{A}$ is such that $\mathbf{A}\begin{pmatrix}-4 & 1\\ 0 & 0\\ 4 & 5\end{pmatrix} = \begin{pmatrix}32 & 24\end{pmatrix}$. Find a possible matrix $\mathbf{A}$.

Question 38. The matrix $\mathbf{A}$ is such that $\begin{pmatrix}2 & -3\\ 0 & 6\end{pmatrix}\mathbf{A} = \begin{pmatrix}13 & 1\\ -6 & -18\end{pmatrix}$. Find a possible matrix $\mathbf{A}$.

Question 39. The matrix $\mathbf{A}$ is such that $\mathbf{A}\begin{pmatrix} -5 & 2 & -4 \\ 10 & 2 & 16 \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$. Find the matrix $\mathbf{A}$.

Question 40. Given $\mathbf{A} = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, show that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 41. Given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$, use the matrices $\mathbf{A}$ and $\mathbf{B}$ to show that matrix multiplication is not, in general, commutative.

Question 42. Given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix}$, show that the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under multiplication.

Question 43. It is given that $\mathbf{A} = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$.

a) Show that $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.

b) Explain what part (a) shows.

Question 44. It is given that $\mathbf{A} = \begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix}$. Show that $(\mathbf{AB})^2 \neq \mathbf{A}^2\mathbf{B}^2$.

Question 45. It is given that $\mathbf{A} = \begin{pmatrix} 5a & -3 \\ 1 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} p \\ q \end{pmatrix}$, where a is a constant. It is also given that $\mathbf{AB} = 4\mathbf{B}$. Find the value of a.

Question 46. It is given that $\mathbf{A} = \begin{pmatrix} a & 1 & 2 \\ 0 & 3 & a \\ a & 2 & 1 \end{pmatrix}$ and $\mathbf{A}^2 = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix}$. Find the value of a.

Question 47. Given $\mathbf{A} = \begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix}$ and $\mathbf{AB} = \mathbf{BA}$, find the value of x.

Question 48. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 49. It is given that $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 50. It is given that $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}$:

a) Find $\mathbf{A}^2$.
b) Find $\mathbf{A}^3$.
c) Find $\mathbf{A}^4$.
d) Write down a general matrix for $\mathbf{A}^k$.

Question 51. Find the values of a and b such that $\begin{pmatrix} 2 & a \\ 2b & 1 \end{pmatrix}\begin{pmatrix} 3a & 4 \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix}$.

Question 52. Find the values of a and b such that $\begin{pmatrix} 3a & -2 \\ 5a & -3 \end{pmatrix}\begin{pmatrix} 8 & -3a \\ 4a & 1 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$.

Question 53. Find the values of a and b such that $\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix}^2 = \mathbf{I}$.

Question 54. Find the values of a and b such that $\begin{pmatrix} 3b & 4 \\ 5a & 2 \end{pmatrix}^2 = \mathbf{0}$.

Question 55. Given $\mathbf{A} = \begin{pmatrix} 2 & 5 \\ 10 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 7 & 2 \\ 4 & 7 \end{pmatrix}$, find the value of p and q given $p\mathbf{A} - \mathbf{I} = q\mathbf{B}$.

Question 56. Given $\mathbf{A} = \begin{pmatrix} 1 & 10 \\ 50 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 3 \\ 15 & 1 \end{pmatrix}$, find the value of p and q given $p\mathbf{A} + q\mathbf{B} = 5\mathbf{I}$.

Question 57. Given $\mathbf{A} = \begin{pmatrix} 5 & 1 \\ x & 2 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 2y & 3z \\ z & y \end{pmatrix}$ and $\mathbf{AB} = \begin{pmatrix} 38 & 123 \\ 52 & 150 \\ 8 & 3 \end{pmatrix}$, find the values of x, y and z.

Question 58. Given $\mathbf{A} = \begin{pmatrix} -1 & x & 3 & 4 \\ x & 3 & y & y \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 4 & 5 \\ 2 & 3 \\ x & y \\ 4 & 2 \end{pmatrix}$ and $\mathbf{AB} = 2\mathbf{I}$, find the values of x and y.

Question 59. It is given that the matrices $\mathbf{A} = \begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix}$ are commutative under multiplication. Find the value of x.

Question 60. It is given that the matrices $\mathbf{A} = \begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix}$, where k is a constant. Find the possible values of k such that $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 61. It is given that $\mathbf{A} = \begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix}$. Show that $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

Question 62. It is given that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{I}$ are 2×2 matrices, where $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$ and $\mathbf{I}$ is the identity matrix. Given also that $4\mathbf{A} - 2\mathbf{I} = \mathbf{AB}$, find the matrix $\mathbf{A}$.

Question 63. It is given that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{I}$ are 2×2 matrices, where $\mathbf{A} = \begin{pmatrix} a+3 & -2a \\ 6-a & a \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} a-6 & a-3 \\ 10-a & a-5 \end{pmatrix}$ and $\mathbf{I}$ is the identity matrix.
Given also that $k\mathbf{I} = \mathbf{AB}$, find the values of a and k.

Question 64. It is given that $\mathbf{A} = \begin{pmatrix} 1 & -1 & -3 \\ 0 & x & 1 \\ 3 & -2 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} \frac{4}{3} & 1 & -\frac{5}{3} \\ 2 & x+13 & 3 \\ 1 & -8 & 0 \end{pmatrix}$.
par These matrices are commutative under multiplication. Find the value of x.

Question 65. Prove that matrix addition is commutative for 2×2 matrices.

Question 66. Prove that matrix subtraction is not commutative for 2×2 matrices.

Question 67. Prove that matrix multiplication is not commutative for 2×2 matrices.

Question 68. In each case, write down the transpose of the given matrix:

a) $\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 6 & -7 \end{pmatrix}$

b) $\mathbf{B} = \begin{pmatrix} 4 & 2 & 0 \\ -9 & 1 & 4 \end{pmatrix}$

c) $\mathbf{C} = \begin{pmatrix} 1 & 2 & -7 \\ 10 & -9 & 2 \\ 3 & 0 & 8 \end{pmatrix}$

Question 69. Given $\mathbf{A} = \begin{pmatrix} 5 & -8 & 5 \\ 2 & 1 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 8 & -5 \\ 2 & -1 \end{pmatrix}$, construct a list of all the possible products of two of $\mathbf{A}$, $\mathbf{A}^T$, $\mathbf{B}$ and $\mathbf{B}^T$. There is no need to find these products.

Question 70. It is given that $\mathbf{A} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}$:

a) Find the value(s) of k such that $\mathbf{A}\mathbf{A}^T = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$.

b) Find the value(s) of k such that $\mathbf{A}^T\mathbf{A} = 32\begin{pmatrix} 1 & 4 \\ 4 & 20 \end{pmatrix}$.

Question 71. Given $\mathbf{A} = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$:

a) Find $\mathbf{A}\mathbf{A}^T$.
b) Find $\mathbf{A}^T\mathbf{A}\mathbf{A}^T$.

Question 72. Given $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix}$, show that $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Question 73. Given $\mathbf{B} = \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{B}^T\mathbf{A}^T = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$, find the matrix $\mathbf{A}$.

Question 74. Given that $\mathbf{A}$ and $\mathbf{B}$ are both 2×2 matrices, prove that $(\mathbf{AB})^T \equiv \mathbf{B}^T\mathbf{A}^T$.

Question 75. Given $\mathbf{A} = \begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}$ and $(\mathbf{AB}^T - \mathbf{BA}^T)^2 = -25\mathbf{I}$, and x and y are integers such that $1 < x < y$, find the value of x and the value of y.

Question 76. Given that $\mathbf{A}$ and $\mathbf{B}$ are both 2×2 matrices, prove that $(\mathbf{A} + \mathbf{B})^T \equiv \mathbf{A}^T + \mathbf{B}^T$.

Question 77. Given that $\mathbf{A}$ is a 2×2 matrix, prove that $(k\mathbf{A})^T \equiv k\mathbf{A}^T$.

Question 78. Given that $\mathbf{A}$ is a non-singular 2×2 matrix, prove that $\det(\mathbf{A}^T) \equiv \det \mathbf{A}$.

Question 79. Given that $\mathbf{A}$ is a non-singular 2×2 matrix, prove that $(\mathbf{A}^{-1})^T \equiv (\mathbf{A}^T)^{-1}$.